

Liveblocks

Documentation Infrastructure Audit Report

Prepared by VeriOps

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Risk: Moderate

Key Metrics

Pages scanned: 782

Report type: Premium Executive Audit

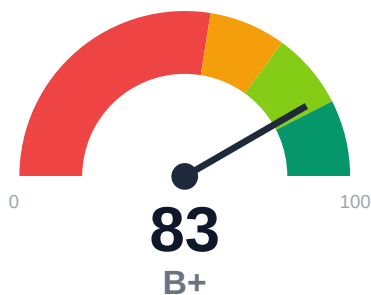
Methodology: 5-pillar automated analysis + expert synthesis

Headline Findings

- Broken internal links: 29
- Link verification inconclusive for 7 URLs (auth/rate-limit/server block); these are excluded from broken-link counts.
- API usage-doc coverage is low: 33.33% (76 API pages detected)
- Freshness visibility is weak: only 40.41% pages show last updated

This report is generated automatically by the VeriOps platform. Estimates are directional.

Executive Summary



Liveblocks documentation audit covered 786 pages with 100% crawl coverage and 83.72% overall confidence. The site contains 29 confirmed broken internal links, all within user-facing documentation. API reference coverage is low at 33.33% across 76 detected API pages. Only 40.41% of pages display last-updated metadata, limiting freshness transparency. Example code reliability estimate stands at 79.72%.

External posture: SEO/GEO issue rate **0.5%**, public API coverage **33.3%**, broken links **29**.

83 Audit Score	B+ Grade	Moderate Risk Band
782 Pages Scanned	29 Broken Links	0.5% SEO Issue Rate

Per-Site Breakdown

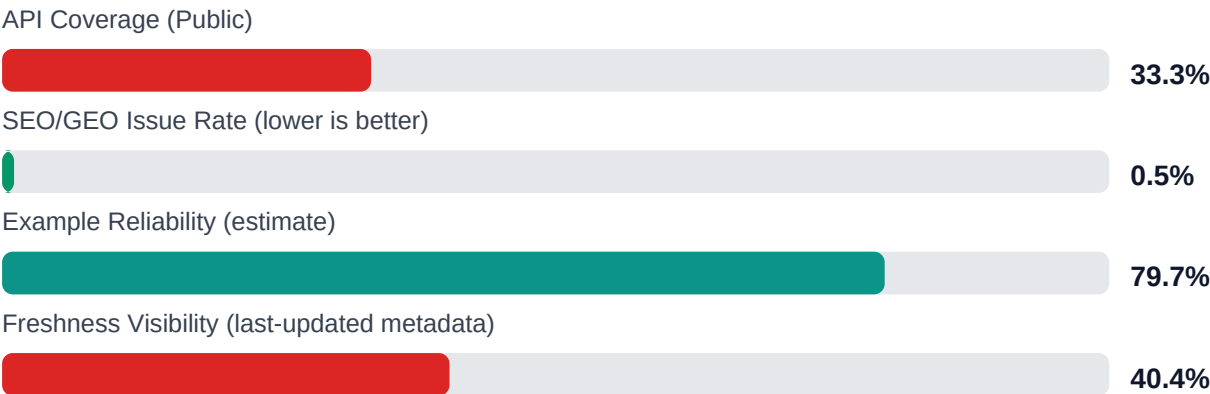
Site URL	Pages	Broken Links	API Cov %	Example Rel %	SEO Issue %
https://liveblocks.io/docs	782	29	33.3	79.7	0.5

Industry Benchmark Comparison

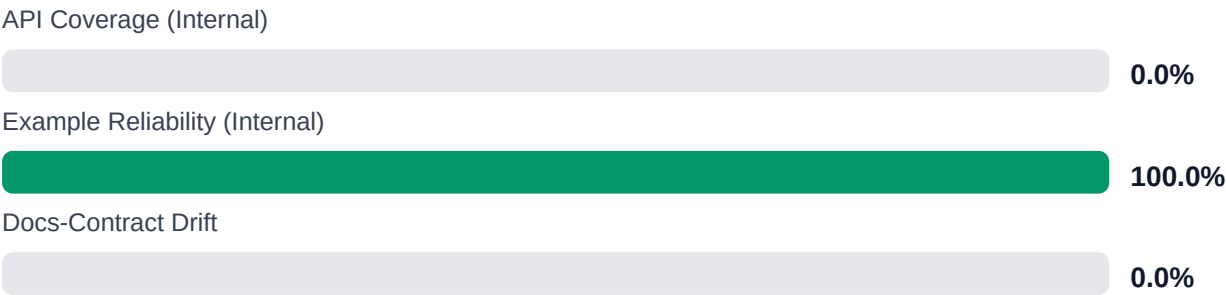
Benchmark	Score	Gap
Top performers (Stripe, Twilio)	92+	-9
Industry average (developer docs)	85	-2
Minimum acceptable	70	+13
Your score	83	-

Gap to industry average: **2 points**. Top-performing developer documentation platforms (Stripe, Twilio, Vercel) score 92+. Closing this gap reduces support ticket volume by an estimated 20-30%.

Board-Level Metrics



Internal Metrics (from scorecard)



Financial Exposure Model

Low Estimate	Base Estimate	High Estimate
\$140,822 annual exposure	\$249,513 annual exposure	\$431,728 annual exposure
Remediation: \$2,090 Monthly loss: \$4,125 Opportunity: \$7,436/mo	Remediation: \$5,445 Monthly loss: \$12,903 Opportunity: \$7,436/mo	Remediation: \$8,800 Monthly loss: \$27,808 Opportunity: \$7,436/mo

Remediation: finding-level effort model | Monthly loss: operational friction model | Opportunity: engineering/support/release delay

Risk Matrix

ID	Issue	Severity	Monthly Loss	Fix Cost
F-LAYERS	Missing required doc layers	HIGH	\$1,782	\$825
F-FRESHNESS	Documentation freshness debt	MEDIUM	\$1,122	\$550
F-RETRIEVAL	RAG retrieval quality risk	MEDIUM	\$2,257	\$990
F-API-COVERAGE	Undocumented API operations	LOW	\$2,904	\$1,100
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	LOW	\$2,244	\$935
F-DRIFT	Code/docs contract drift	LOW	\$1,901	\$660
F-TERMINOLOGY	Terminology inconsistency	LOW	\$693	\$385

Priority Action Plan (Next 14 Days)

ID	Finding	Severity
F-LAYERS	Missing required doc layers	HIGH
F-FRESHNESS	Documentation freshness debt	MEDIUM
F-RETRIEVAL	RAG retrieval quality risk	MEDIUM
F-API-COVERAGE	Undocumented API operations	LOW
F-EXAMPLES-RELIABILITY	Code examples fail or are non-runnable	LOW

Recommended Actions

- Fix all 29 confirmed broken internal documentation links [impact: critical, effort: low]
- Expand API reference coverage from 33.33% to minimum 80% by documenting missing endpoints and improving interlinking [impact: critical, effort: high]

3. Implement last-updated timestamps on all pages to increase freshness coverage from 40.41% to 95%+ [impact: high, effort: low]
4. Audit and update code examples to improve reliability from 79.72% to 90%+ through automated testing [impact: high, effort: medium]
5. Investigate and resolve 7 unverified links blocked by auth/rate-limits to eliminate verification gaps [impact: medium, effort: medium]

Expert Analysis: Strengths and Risks

Strengths

- + Complete crawl coverage at 100% (786 of 786 pages in scope)
- + No repository navigation link breakage (0 repo_broken_links)
- + High overall audit confidence at 83.72%
- + Single-product topology provides clear documentation structure
- + API pages successfully detected (76 pages, no detection limitations flagged)

Risks

- 29 confirmed broken internal links in user-facing documentation degrade navigation and user trust
- API reference coverage at 33.33% leaves two-thirds of API surface undocumented or poorly linked
- 7 unverified links due to auth/rate-limit/server blocks create uncertainty in link health
- Only 40.41% freshness coverage prevents users from assessing content currency
- Example reliability at 79.72% means approximately 1 in 5 code samples may be outdated or broken
- Link verification confidence at 75.56% is lowest among audit dimensions
- SEO/geo issue rate at 0.54% may impact discoverability in certain markets

Limitations

- * External audit cannot verify content accuracy, only structural and link integrity
- * 7 links remain unverified due to authentication or rate-limiting; actual breakage unknown
- * API coverage measurement depends on detection heuristics; actual coverage may vary if non-standard page structures exist
- * Example reliability is estimated through static analysis; runtime execution success not guaranteed
- * Audit cannot assess documentation completeness, clarity, or alignment with actual product behavior

Methodology

This audit applies a rigorous, repeatable methodology combining automated analysis with expert synthesis. The platform evaluates documentation quality across five pillars.

1. Automated Crawl + Structural HTML Analysis

Every public documentation page is crawled and analyzed for structural integrity, link health, navigation depth, and content organization. The crawler detects broken links, orphaned pages, redirect chains, and missing metadata.

2. SEO/GEO Scoring Model (24 Checks)

Each page is evaluated against 8 GEO checks (LLM/AI search optimization) and 16 SEO checks (traditional search engine optimization). Checks cover meta descriptions, heading hierarchy, fact density, URL structure, internal linking, and content depth.

3. API Reference Cross-Coverage Analysis

The platform compares published API reference documentation against the actual API surface (OpenAPI specs, GraphQL schemas, gRPC protos). Coverage gaps, undocumented endpoints, and stale references are identified.

4. Code Example Reliability Estimation

Code examples embedded in documentation are analyzed for syntactic correctness, completeness, and consistency with current API signatures. Reliability scores estimate the likelihood of examples working as documented.

5. Executive Synthesis and Prioritization

All quantitative findings are synthesized into actionable executive narratives. The analysis identifies patterns, prioritizes risks, and generates strategic recommendations calibrated to business impact.

Appendix A -- Economic Model Assumptions

Assumption	Value
engineer_hourly_usd	110.0
support_hourly_usd	50.0
release_count_per_month	8.0
baseline_manual_sync_hours_per_week	12.0
avg_release_delay_hours	4.5
monthly_support_tickets	800.0
docs_related_ticket_share	0.22
avg_ticket_handling_minutes	24.0

Note: estimates are directional and should be calibrated with client rates, release cadence, and support volume.

Appendix B -- Evidence Traceability

Finding	Evidence Source	Confidence
F-LAYERS -- Missing required doc layers	policy pack + frontmatter layer scan	HIGH
F-FRESHNESS -- Documentation freshness debt	frontmatter date scan	HIGH
F-RETRIEVAL -- RAG retrieval quality risk	retrieval_evals_report.json	HIGH
F-API-COVERAGE -- Undocumented API operations	OpenAPI + docs text scan	HIGH
F-EXAMPLES-RELIABILITY -- Code examples fail or are non-runnable	examples_smoke_report.json	HIGH
F-DRIFT -- Code/docs contract drift	pr_docs_contract.json + api_sdk_drift_report.json	HIGH
F-TERMINOLOGY -- Terminology inconsistency	glossary.yml forbidden terms scan	HIGH

All findings are traceable to automated checks. Evidence sources reference specific crawl data, API specs, or code analysis results.